



R.M.D ENGINEERING COLLEGE (An Autonomous Institution) Chennai - 601206.

Department of Computer Science and Engineering

MDR SENTINEL Hospital Outbreak Intelligence

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Abstract

Healthcare facilities face significant challenges in controlling multidrug-resistant (MDR) infections due to delayed manual contact tracing and fragmented data systems. Existing infection-control approaches are largely reactive, identifying outbreaks only after laboratory confirmation. This project introduces an AI-driven infection intelligence platform for real-time outbreak prevention. A hospital digital twin is created using BLE-based indoor tracking to model real-time movement of patients and staff. Graph-based analytics identify hidden exposure chains and potential super-spreader events instantly. An AI-powered MDR risk prediction engine forecasts outbreaks before they occur. The system integrates EHR, lab, and sensor data into a unified surveillance framework. A real-time alert dashboard delivers actionable notifications within seconds. An Infection Replay Engine enables outbreak playback and root-cause analysis. This solution enables proactive, data-driven infection control with improved safety and compliance.



Introduction

Multi-Drug Resistant (MDR) infections have presented a critical problem in contemporary healthcare, resulting in extended hospital stays, increased healthcare expenses, and escalating mortality rates. The majority of hospitals rely on traditional and time-consuming contact tracing, delayed test results, and fragmented data systems, which makes infection control slow and passive.

The project suggests an AI-powered infection intelligence system, which will allow hospitals to monitor and control infections in real time and prevent outbreaks. The system will utilize technology such as Bluetooth Low Energy, Internet of Things, and digital twin modelling to monitor the movements and interactions of patients and healthcare staff within the healthcare setting.

The system will utilize Electronic Health Records, test results, and sensor data to provide a unified infection control system. The system will utilize graph-based analytics and machine learning to identify exposure patterns, high-risk contacts, and potential outbreaks. The system will include features such as infection alerts and infection replays, which will enhance effective decision-making. The system will revolutionize traditional infection control by providing a data-driven approach, which will enhance efficiency and safety.



Literature Survey

Title of The paper	Description	Publication details
Estimating Potential Infection Transmission Routes in Hospital Wards Using Wearable Proximity Sensors	Contacts between patients, patients and health care workers (HCWs) and among HCWs represent one of the important routes of transmission of hospital-acquired infections (HAI). A detailed description and quantification of contacts in hospitals provides key information for HAIs epidemiology and for the design and validation of control measures.	R. E. Ugandar., October 3, 2023
Contact tracing & super-spreaders in the branching-process model	This paper analyzes how super-spreader events influence the spread of airborne infections and the effectiveness of contact tracing. It shows that even when many people are infected in single events, efficient contact tracing with high detection rates can still help control disease transmission.	Johannes Müller, Volker Hösel (2023)



Existing System vs Our System

Aspect	Existing System	Our System – MDR Sentinel
Monitoring Method	Relies on manual observation and delayed reporting by hospital staff.	Uses IoT sensors and BLE tags to continuously track movement and interactions in real time.
Contact Tracing	Manual contact tracing which may take days or weeks and is often incomplete.	Automated graph-based contact tracing detects exposure chains instantly and accurately.
Outbreak Detection	Identifies infections only after symptoms appear or lab reports confirm cases.	AI-driven predictive analytics forecasts infection risks before outbreaks occur.
Data Integration	Hospital data such as EHR, lab reports, and movement records are stored separately.	Integrates multiple data sources into a unified platform for intelligent infection monitoring.
Decision Support	Limited insights, making it difficult for infection control teams to act quickly.	Provides real-time dashboards, heatmaps, alerts, and analytics for faster data-driven decisions.



PROPOSED SYSTEM

- Real-Time Monitoring System:** Develop an AI-driven platform that continuously monitors the movement of hospital staff, patients, and equipment using BLE tags and IoT sensors.
- Automated Contact Tracing:** Implement a smart contact tracing system that automatically records interactions between individuals and identifies possible infection exposure events.
- Centralized Data Integration:** Integrate hospital data sources such as Electronic Health Records (EHR), lab reports, and environmental sensor data into a unified platform.
- Risk Scoring Engine:** Design a dynamic risk scoring mechanism that evaluates infection probability based on contact duration, proximity, and environmental conditions.
- Graph-Based Infection Analysis:** Use graph algorithms to map interaction networks and identify high-risk individuals or potential super-spreaders.
- Predictive Outbreak Detection:** Apply machine learning models to analyze historical data and predict potential infection outbreaks in advance.
- Hotspot Visualization:** Generate heatmaps and digital twin dashboards to visually display infection risk zones and contamination hotspots within the hospital.
- Real-Time Alert System:** Provide instant notifications to healthcare staff when they enter high-risk areas or come into contact with potentially infected individuals.
- Decision Support Dashboard:** Offer interactive dashboards and reports to help hospital administrators and infection control teams take timely preventive actions.



METHODOLOGY

- Data Collection:** BLE tags and IoT sensors collect real-time location and interaction data of doctors, nurses, patients, and medical equipment within the hospital.
- Data Integration:** The collected data is securely transmitted and integrated with hospital systems such as Electronic Health Records (EHR), lab reports, and staff information databases.
- Contact Detection:** The system identifies interactions between individuals based on proximity and duration to detect possible exposure events.
- Risk Analysis:** A risk scoring algorithm evaluates infection probability using factors such as contact duration, distance, and environmental conditions.
- Graph-Based Contact Tracing:** Graph network analysis maps relationships between individuals to identify high-contact nodes and potential super-spreaders.
- Predictive Analysis:** Machine learning models analyze historical infection patterns to predict outbreak risks and generate early warnings.
- Visualization & Alerts:** Results are displayed on a digital dashboard with heatmaps and real-time alerts to help hospital staff take immediate preventive actions.



Algorithm/Techniques/Tools Used

The MDR Sentinel system uses a combination of advanced algorithms, IoT technologies, and AI models to monitor and prevent hospital infection outbreaks. BLE-based trilateration algorithms are used to track the real-time location of staff, patients, and equipment within the hospital environment. Graph network analysis algorithms such as centrality and PageRank are applied to identify high-contact individuals and potential super-spreaders in the contact network. Machine learning models including Random Forest and LSTM are used to predict infection risks based on historical and environmental data. A dynamic risk scoring algorithm calculates infection probability using factors such as distance, duration of contact, and contamination level. Heatmap generation techniques visualize germ trails and contamination hotspots across hospital zones. Real-time alerting mechanisms use proximity detection to notify healthcare staff when entering high-risk areas. WebSocket-based communication enables live data streaming to monitoring dashboards. The system also integrates hospital data sources such as EHR records, lab reports, and IoT sensor inputs. These combined techniques enable proactive infection surveillance and faster decision-making in healthcare facilities.



SETUP REQUIREMENTS

HARDWARE REQUIREMENTS:

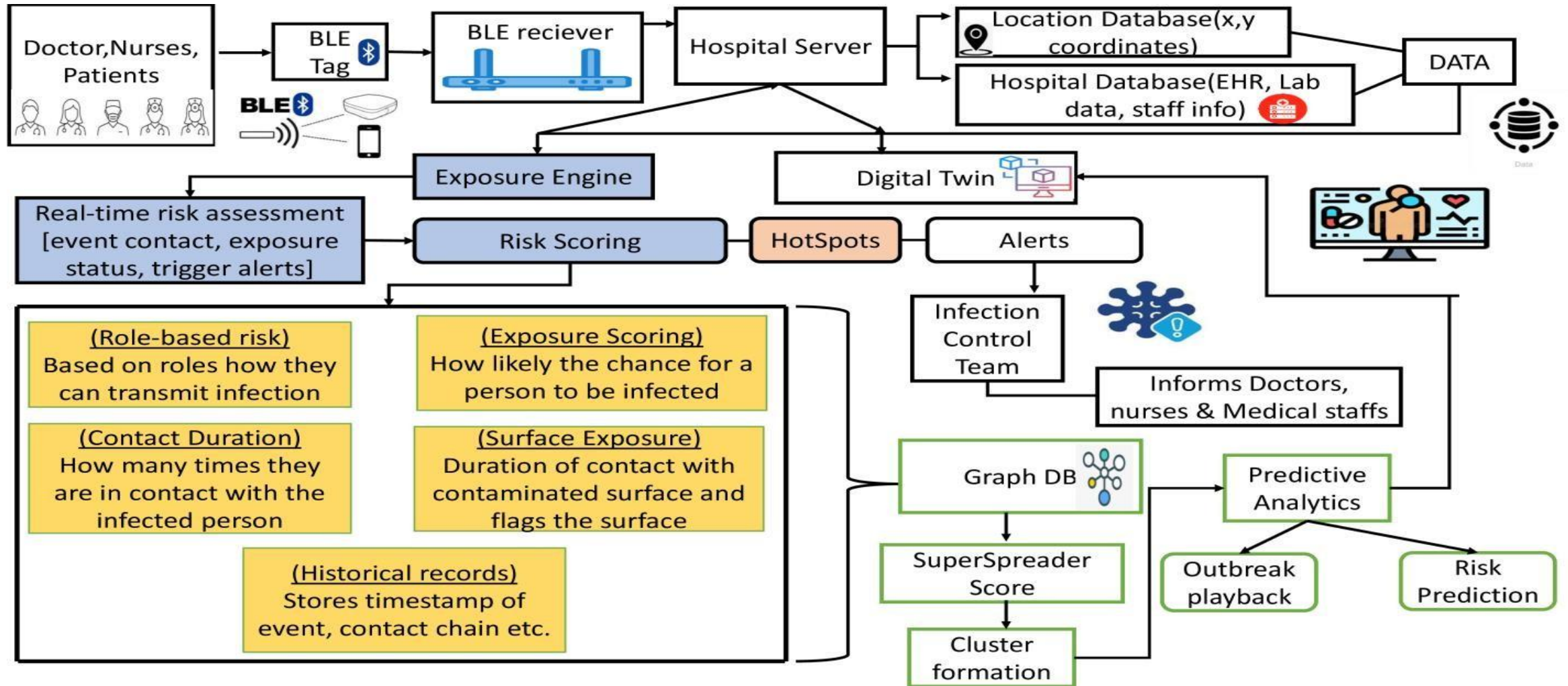
- **BLE Tags / Badges (for patients and staff tracking)**
- **BLE Receivers / Sensors (installed across hospital zones)**
- **Wi-Fi Router / Network Infrastructure**
- **Server System (Local Server / Cloud Access Device)**
- **Computers / Monitoring Systems (for dashboard access)**
- **Mobile Devices (for alerts and notifications)**
- **Power Supply Units**

SOFTWARE REQUIREMENTS:

- **Operating System: Linux (Ubuntu 22.04 LTS) or Windows**
- **Frontend: HTML, CSS, JavaScript, React.js**
- **Backend: Node.js (Express.js)**
- **Database: Neo4j (Graph Database), PostgreSQL / MongoDB**
- **AI/ML Models: Python (Scikit-learn, TensorFlow / PyTorch)**
- **IoT Communication: MQTT / WebSockets**
- **Visualization: Digital Twin Dashboard (2D/3D Web-based Interface)**
- **APIs & Integration: FHIR APIs / Hospital EHR Integration**
- **Cloud Platform: AWS / Firebase / Azure**



ARCHITECTURE DIAGRAM



Financial Model

COST:

- Initial investment includes sensor deployment (₹1.5–2 Lakhs per hospital), cloud infrastructure, and system integration

REVENUE STREAMS:

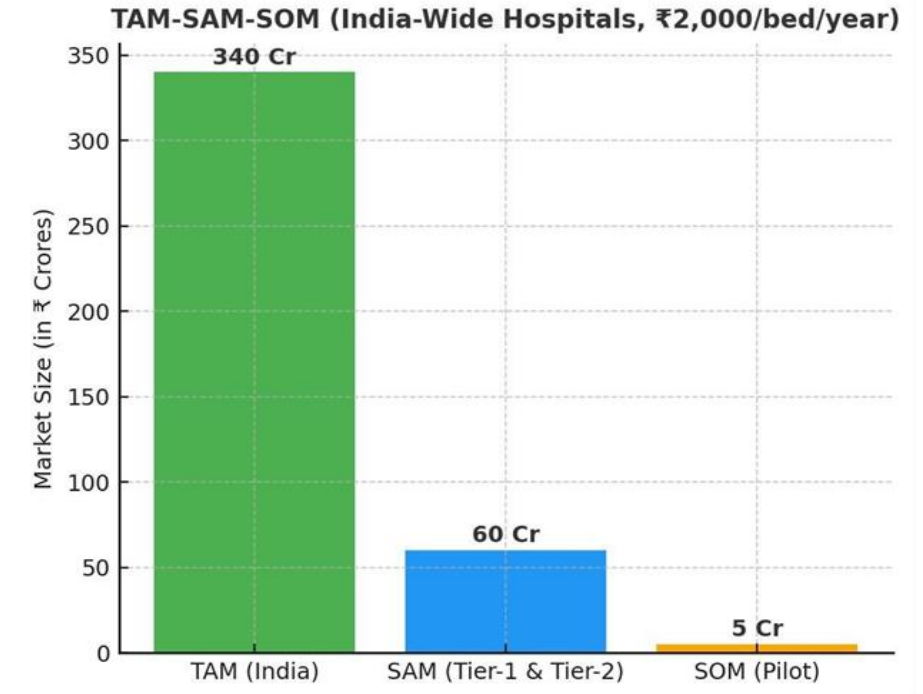
- One-time setup fee: ₹2–3 Lakhs per hospital
- Monthly SaaS subscription: ₹40K–₹1L
- Additional income from paid compliance and analytics reports

Financial Outlook:

- Scalable model with estimated break-even in 12–18 months

MARKET EXPANSION STRATEGY:

- Distribution through direct hospital onboarding, government tenders, and healthcare partnerships
- Scaling via state health networks and centralized cloud deployment across multiple hospitals



REVENUE (TAM–SAM–SOM)

- **TAM (POTENTIAL MARKET) → ~₹340 CR**
- **SAM (NEAR-TERM TARGET) → ~₹60 CR**
- **SOM (PILOT DEPLOYMENT REVENUE) → ~₹5 CR**



Expected Outcomes

- **Accurate Biomedical Waste Segregation** into Dry, Wet, Metal, and Hazardous types.
- **Real-time Waste Tracking** with suggestions to improve hospital sustainability.
- **Digital Link with Recycling Centers** and automated SDG-based PDF waste reports.
- **Awareness & Engagement** through SDG-based games, quizzes, and rewards.
- **Community and Event Participation** to build a global sustainability network.
- **Future Integration with Smart Bins** for automated sorting and IoT monitoring.
- **Support for Healthcare SDGs** promoting safer environments and responsible waste management.



THANK YOU